

Flexible protections can be configured thanks to option bytes:

- Readout protection (RDP) to protect the whole memory. Three levels are available:
 - Level 0: no readout protection
 - Level 1: memory readout protection: the flash memory cannot be read from or written to if either debug features are connected, boot in RAM or bootloader is selected
 - Level 2: chip readout protection: debug features (Cortex-M0+ serial wire), boot in RAM and bootloader selection are disabled. This selection is irreversible.

Table 3. Access status versus readout protection level and execution modes

Area	Protection level	User execution			Debug, boot from RAM or boot from system memory (loader)		
		Read	Write	Erase	Read	Write	Erase
Main memory	1	Yes	Yes	Yes	No	No	No
	2	Yes	Yes	Yes	N/A	N/A	N/A
System memory	1	Yes	No	No	Yes	No	No
	2	Yes	No	No	N/A	N/A	N/A
Option bytes	1	Yes	Yes	Yes	Yes	Yes	Yes
	2	Yes	No	No	N/A	N/A	N/A

- Write protection (WRP): the protected area is protected against erasing and programming. Two areas per bank can be selected, with 2-Kbyte granularity.
- Proprietary code readout protection (PCROP): a part of the Flash memory can be protected against read and write from third parties. The protected area is execute-only: it can only be reached by the STM32 CPU as instruction code, while all other accesses (DMA, debug and CPU data read, write and erase) are strictly prohibited. An additional option bit (PCROP_RDP) determines whether the PCROP area is erased or not when the RDP protection is changed from Level 1 to Level 0.

3.3.1 Securable area

A part of the Flash memory can be hidden from the application once the code it contains is executed. As soon as the write-once SEC_PROT bit is set, the securable memory cannot be accessed until the system resets. The securable area generally contains the secure boot code to execute only once at boot. This helps to isolate secret code from untrusted application code.

3.4 Embedded SRAM

STM32C011x4/x6 devices have 6 Kbytes of embedded SRAM with parity. Hardware parity check allows memory data errors to be detected, which contributes to increasing functional safety of applications.

The memory can be read/write-accessed at CPU clock speed, with 0 wait states.